Physics of the Earth and Planetary Interiors, 62 (1990) 109–125
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

A new kinematic model for Iberia; further palaeomagnetic and isotopic age evidence

K.M. Storetvedt [1]*, J.G. Mitchell [1], M.C. Abranches [2] and S. Oftedahl [3]

[1] *Department of Physics, University of Newcastle upon Tyne, Newcastle upon Tyne NE1 7RU (U.K.)*
[2] *Departamento de Fisica, Faculdade de Ciencias, Universidade de Lisboa, P-1200 Lisboa (Portugal)*
[3] *Geofysisk institutt, Universitetet i Bergen, N-5007 Bergen (Norway)*

(Received April 26, 1989; revision accepted May 16, 1989)

ABSTRACT

Storetvedt, K.M., Mitchell, J.G., Abranches, M.C. and Oftedahl, S., 1990. A new kinematic model for Iberia; further palaeomagnetic and isotopic age evidence. Phys. Earth Planet. Inter., 62: 109–125.

Palaeomagnetic results and K–Ar age data from the Algarve Igneous Province have uncovered three ancient relative field components. It is inferred that the principal magnetization of the Monchique Complex, with a mean direction of $D = -42°$, $I = 181°$, ($\alpha_{95} = 4.5°$), was acquired at around the time of the pluton's emplacement some 72 ± 2 Ma ago. A second magmatic pulse affected the region at 68 ± 2 Ma, leaving behind a relative field component of $D = 42°$, $I = 341°$, ($\alpha_{95} = 1.7°$). Furthermore, an overprinted magnetization, characterized by shallow inclination and somewhat smeared northwesterly–southeasterly declination, most likely originates from the Lower Tertiary. The new palaeomagnetic data are seen in conjunction with previously published Upper Cretaceous results from the Sintra and Sines intrusive complexes. The resulting database gives entirely new perspectives for evaluating the plate tectonic history of the Peninsula in Upper Cretaceous–Lower Tertiary time. A two-phase rotation of the Iberian micro-plate, an anticlockwise one of ~ 40° followed by a clockwise one of ~ 70°, is clearly in evidence. The net clockwise rotation provided a major impetus for the compressive deformation in the Pyrenees and the Bay of Biscay, and led to extensive remagnetization of older rock formations in the northern regions of Iberia. Previous palaeomagnetic conclusions, in favour of a single anticlockwise rotation of ~ 35°, supposed to be directly connected with the formation of the Bay of Biscay, appear to be artefacts which stem from an incorrectly adopted palaeomagnetic reference frame.

1. Introduction

Recent palaeomagnetic results and isotopic age data for late Cretaceous intrusions of Portugal (Storetvedt et al., 1987) suggested that the Upper Cretaceous was a period of major regional plate tectonic activity, possibly including both clockwise and anticlockwise rotations of the Iberian Peninsula. To elucidate further this intriguing tectonic question, additional data are reported here from other members of the alkaline igneous province of Portugal: The Monchique Syenite Complex, and two minor intrusions from the littoral Algarve. The results fit into the palaeomagnetic/age pattern previously obtained, giving the new plate tectonic model for Iberia a firm basis.

The Monchique pluton, covering a surface area of 63 km^2 with an east–west elongate shape, is a nepheline syenite intrusion comprising a variety of feldspathoidal rock types in association with a larger number of mostly narrow dykes of variable composition (see below). The Complex is intrusive into Carboniferous argillites and at Serra de Monchique, for example, a hornfelsic thermal aureole is very well exposed. Along the coastal strip SSW of the Monchique pluton, the Triassic–

* On leave from: Geofysisk institutt, Universitetet i Bergen, N-5007 Bergen, Norway.

0031-9201/90/$03.50 © 1990 – Elsevier Science Publishers B.V.

Jurassic sediments of the Algarve Basin have been invaded by intrusions in the form of bosses, dykes and sills of basanites, limburgites, ankaramites, etc. (Assunçao, 1940) which show petrological affinities with the Monchique Complex (Rock, 1982). According to Rock the main complex consists of (1) 95% nepheline syenites (foyaites and pulaskites), (2) 4.5% breccias, and (3) 0.5% basic rock segregations. A total of about 300 dykes (cutting the syenites) have been recorded, of which 50% are lamprophyres (monchiquites, camptonites, etc.), 30% have felsic composition (trachytes, phonolites, etc.), 15% are basaltic varieties, and 5% miscellaneous types (pegmatites, aplites, etc.). Figure 1 shows the geographic locations of the alkaline intrusions investigated and/or discussed in the present paper (a), together with a geological sketch map of the Monchique Complex (b), simplified after Goncalves (1967).

2. Sampling and laboratory techniques

Samples for the present study were obtained by drilling and by hand-sampling. The Salema intrusion is particularly accessible in a major quarry which provides some 100 m of exposure. From this extensive outcrop, a total of 20 samples (from three different sites) were collected. Apart from pronounced secondary alteration associated with some prominent fractures, the Salema material has a macroscopically very fresh appearance. In contrast, the Burgau intrusion is very poorly exposed and only one relatively fresh hand-sample was obtained.

The Monchique Massif was sampled at 19 sites, of which 10 were nepheline syenites (nos. 1–6 and 8–11), three sites were in mafic segregations (nos. 12, 13 and 15), one site was in the hornfelsic thermal aureole (no. 14), and the remaining five sampling locations (nos. 7 and 16–19) were in thin lamprophyre dykes.

In the laboratory, cylindrical specimens with diameter and height of 1.9 cm were cut for palaeomagnetic investigations. Remanent magnetization was measured using a Digico spinner magnetometer, a Molspin magnetometer, and a Cryogenic magnetometer (for the hornfelsic sediment). Alternating field (AF) demagnetization was carried out using a two-axis tumbler, and a Schonsted TM-1 was used for thermal demagnetization.

The K–Ar ages were in all cases based on 'whole-rock' analysis (Wilkinson et al., 1986).

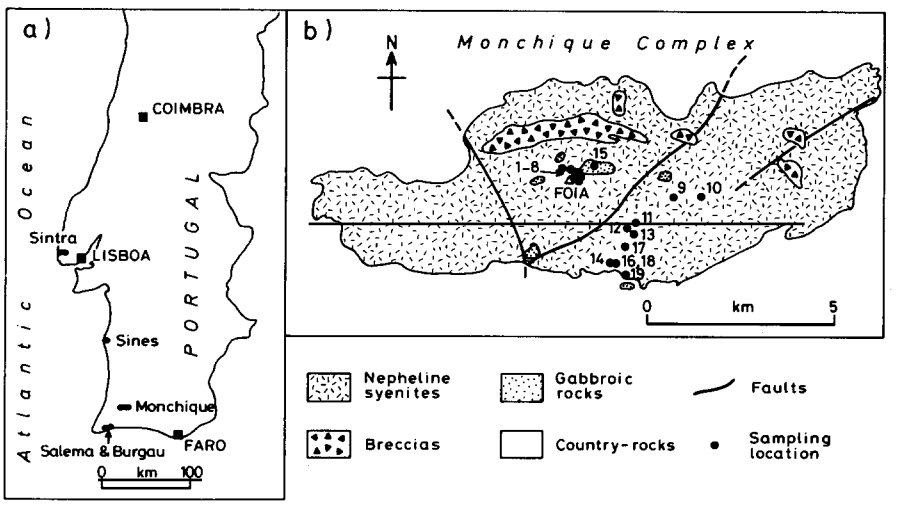


Fig. 1. (a) Sketch map showing the geographic locations of Monchique Syenite Complex, and littoral Algarve basic intrusions at Salema (S) and Burgau (B). Two other representatives of the Alkaline Igneous Province of Portugal, the Sintra and Sines complexes, are also marked. (b) Geological sketch map of the Monchique pluton, simplified after Goncalves (1967), including locations of palaeomagnetic sampling sites (numbered).

These analyses were performed by crushing the samples and taking the 20–40 size fraction to homogenize the material before argon analysis. A portion of this material was then reduced to about 200 mesh size for potassium analysis. Potassium contents were determined by flame photometry on an EEL 450 instrument, using a lithium internal standard. Argon isotopic analysis were performed by isotope dilution on a VG 1200 C mass spectrometer, using an ^{38}Ar 'spike'. Gas extractions were by RF induction heating and cleaning with titanium getters, the extraction system being coupled directly to the mass spectrometer.

3. Properties of the magnetic record

All directions of magnetization reported in the present paper have been established by recognition of linear patterns in orthogonal vector projections (Zijderveld, 1967). Whereas material from the Burgau and Salema intrusives has a very simple magnetization structure, the Monchique Complex is a classical example of multiphase magnetization. As is commonly encountered in such cases, parts of the Monchique collection had a remanence composition in which the stability spectra of individual magnetizations were too close for successful component separation. Also, the terminal directions (upon demagnetization) are sometimes associated with very low magnetic moments so that determination of their most stable magnetizations was obscured by experimental noise and viscosity effects. In addition, site 13 (a basic medium-grained rock) has an excessive remanence intensity which is thought to have been acquired by lightning, and site 14 (hornfelsic sediment) was too weakly magnetic for measurement. In all, the palaeomagnetic database for the Monchique Massif includes only 11 of the 19 sites which were collected, but all main rock types (nepheline syenites, mafic segregations, and lamphrophyre dykes) are represented.

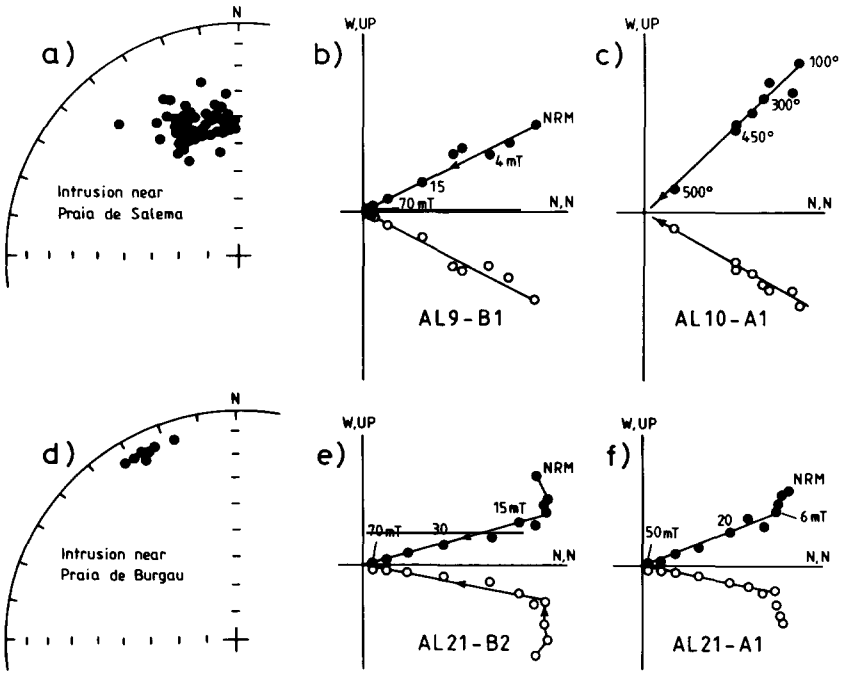


Fig. 2. Populations of characteristic remanence directions for the Salema (a) and Burgau (d) intrusions. Orthogonal vector plots (Zijderveld, 1967) display representative examples; diagrams (b) and (c) depict Salema results, and (e) and (f) display Burgau data. Projection is equal area, and by convention a lower (upper) hemisphere direction plots as a closed (open) symbol. In the vector diagrams projections in the vertical and horizontal planes are marked by open and closed symbols respectively.

3.1. *Salema and Burgau intrusions*

Apart from a modest low-stability component with fairly steep inclinations (probably of viscous origin) the two rock bodies possess very distinct groupings of magnetization directions. The Salema results ($N = 70$) are grouped around a mean directions of $D = +42°$, $I = 341°$, $\alpha_{95} = 1.7°$, whereas the Burgau intrusions (limited to $N = 7$, the total number of specimens available for study) form a shallower but consistent grouping with a mean of $D = +11°$, $I = 333°$, $\alpha_{95} = 2.3°$. Figure 2 displays the two magnetization populations, with some representative demagnetization results for individual samples.

3.2. *Monchique Complex*

Figure 3 depicts the distribution of all of the component directions obtained, among which three groupings, A, B and C, can be distinguished. The

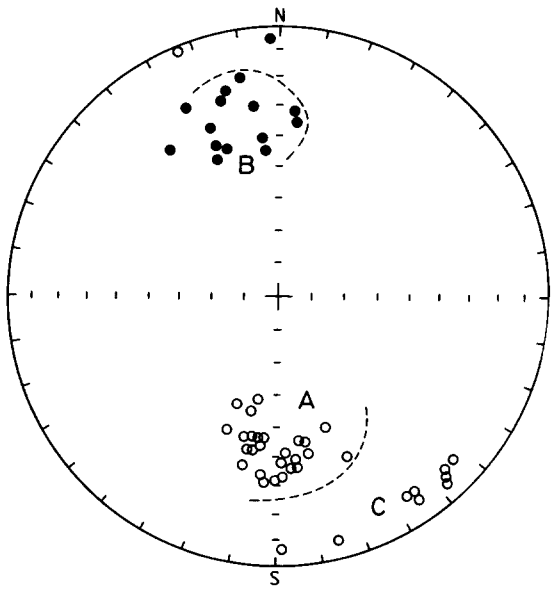


Fig. 3. Distribution of characteristic remanence directions recovered from samples of the Monchique Complex. The A magnetization is interpreted as the original component whereas B and C are inferred to be palaeomagnetic overprints; the B group agrees with the magnetization of the Salema intrusion, and the C group agrees with the Burgau results. The spread in declination for the C magnetization is associated with a relative axis shift. Conventions as for Fig. 2.

A component, the principal magnetization of the pluton, has a mean direction of $D = -42°$, $I = 182°$, $\alpha_{95} = 4.5°$, which is not significantly different from that of a previous study ($D = -37°$, $I = 182°$, $\alpha_{95} = 6.5°$; Van der Voo, 1969). The B direction is indistinguishable from that of the Salema magnetization, and the C component agrees with the Burgau magnetization axis (see also below).

At both site and specimen level the various Monchique remanence components occur in a variety of component combinations. Figure 4 shows results from samples that are dominated by the reverse A magnetization. Overprinting (see below) by a magnetization of normal polarity, generally the B magnetization or occasionally the Present Earth's Field (PEF) component, is exemplified for three samples, numbers. 35-d1, 62-A and 62-C. Sample 35-d1 has a well-defined northerly-trending remagnetization circle, which is consistent with the presence of the NNW-directed B remanence. An excessively strong overlap of the combined N + R component spectra after 25 mT treatment, and the onset of an erratic stage after 65 mT, have, however, prevented the experimental demonstration of the B component.

Figure 5 depicts further examples of coexisting A and B magnetizations. For specimens 1-A, 21-A2 and 23-A2 the B magnetization carries the bulk of the magnetic moment, and in such cases an erratic/unstable state generally prevents the proper definition of the end-point magnetization (component A). Sample 21-A2 is an exception in that both components are experimentally resolved. Specimen 35-B2 (d and f) provides an interesting example of interleaved multicomponent spectra:

(1) the reversed A magnetization is erased in two blocking temperature ranges, NRM–500 °C (A1) and 580–640 °C (A2);

(2) the existence of a normally directed magnetization (probably the B component) can be envisaged from the northward-trending remagnetization circle (NRM–500 °C) but too close component overlap with the reverse A magnetization has prevented it from being analytically or experimentally unfolded;

(3) a third magnetization, probably the reverse C remanence (see also below), seems to play an

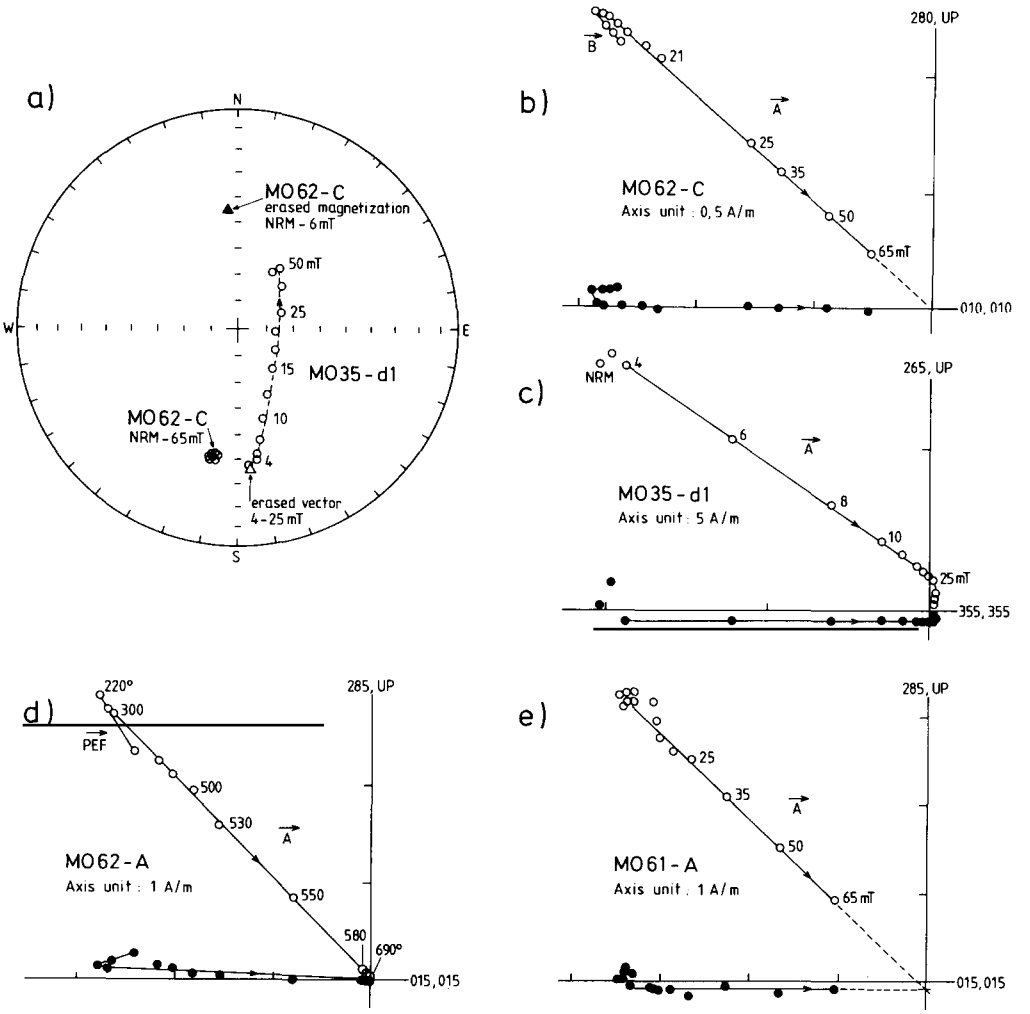


Fig. 4. Demagnetization data for Monchique samples for which the A component predominates. The presence of the B magnetization (a and b), or a Present Earth's Field (PEF) component (e) is very subordinate. Triangles (a) denote directions of magnetization erased over the demagnetization range specified, corresponding to linear segments in associated vector plots (b and c). The orthogonal vector diagrams are closely optimal. General conventions as for Fig. 2.

important role at the higher temperature region of haematite.

Figure 6 displays the C remanence, either as a single magnetization or associated with the B component. Whereas sample 79-A (a and c) tends to display single-component C behaviour, the other examples correspond to three-component situations. Numbers 9-A1 and 12-A2, for example, have two C vectors (C1 and C2), differing only in declination, in addition to a high-temperature possible B magnetization (not properly unfolded). In both of these cases (1) the C1 magnetization (NRM–540 °C) carries the bulk of the remanence, (2) the C2 vector is erased in the 560–630 °C range, and (3) a change in directional trends occurs between 540 and 560 °C. In both cases, the C2 direction coincides with the intersection of the two remagnetization circles.

The Monchique directions outlined in Fig. 3 comprise six nepheline syenite locations (sites 1–4, 8 and 9), two medium-grained mafic rocks (sites 12 and 15), and three lamprophyre dykes (sites 7,

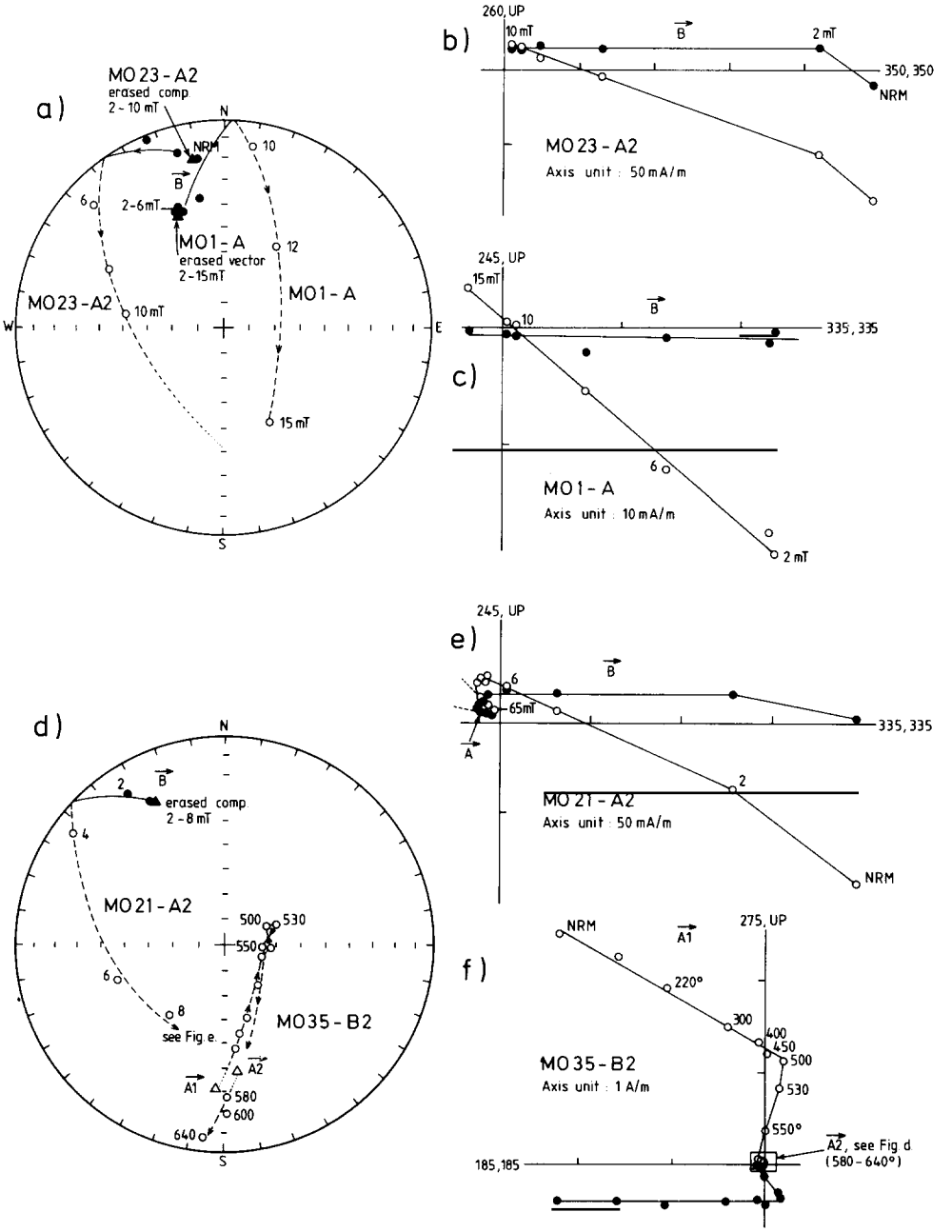


Fig. 5. Further Monchique results with particular emphasis on samples which show relatively strong B component involvement. Sample MO35-B2 (d and f) has its A magnetization erased over two different blocking temperature (T_B) ranges, vector A1 (NRM–500 °C) and A2 (T_B > 580 °C), whereas the normal magnetization present (most likely a B component), which occupies a narrow blocking temperature window around 550 °C, has (due to component overlap) escaped adequate experimental unfolding. Therefore, the 530–580 °C vector segments (e) do not represent a true palaeomagnetic component. Conventions as for Figs. 2 and 4.

17 and 19). Though individual sites can show a preponderance of one of the three characteristic magnetizations (for example, site 2 is dominated by the C component and site 7 by the A magnetization), the general pattern is the coexistence of the three remanences. The C magnetization is observed less frequently than are the other two components. Neither the A nor the B component exhibits dual-polarity structure. The A magnetization has a mean direction striking nearly due south–north, whereas the majority of B group directions have north-northwesterly declinations (see Fig. 3) coinciding with the Salema magnetization (Fig. 2). Conversely, the C remanence (easily distinguishable by its shallow inclinations) has both normal and reverse sub-components. Both magnetite and haematite occur as remanence-carrying minerals responsible for the various

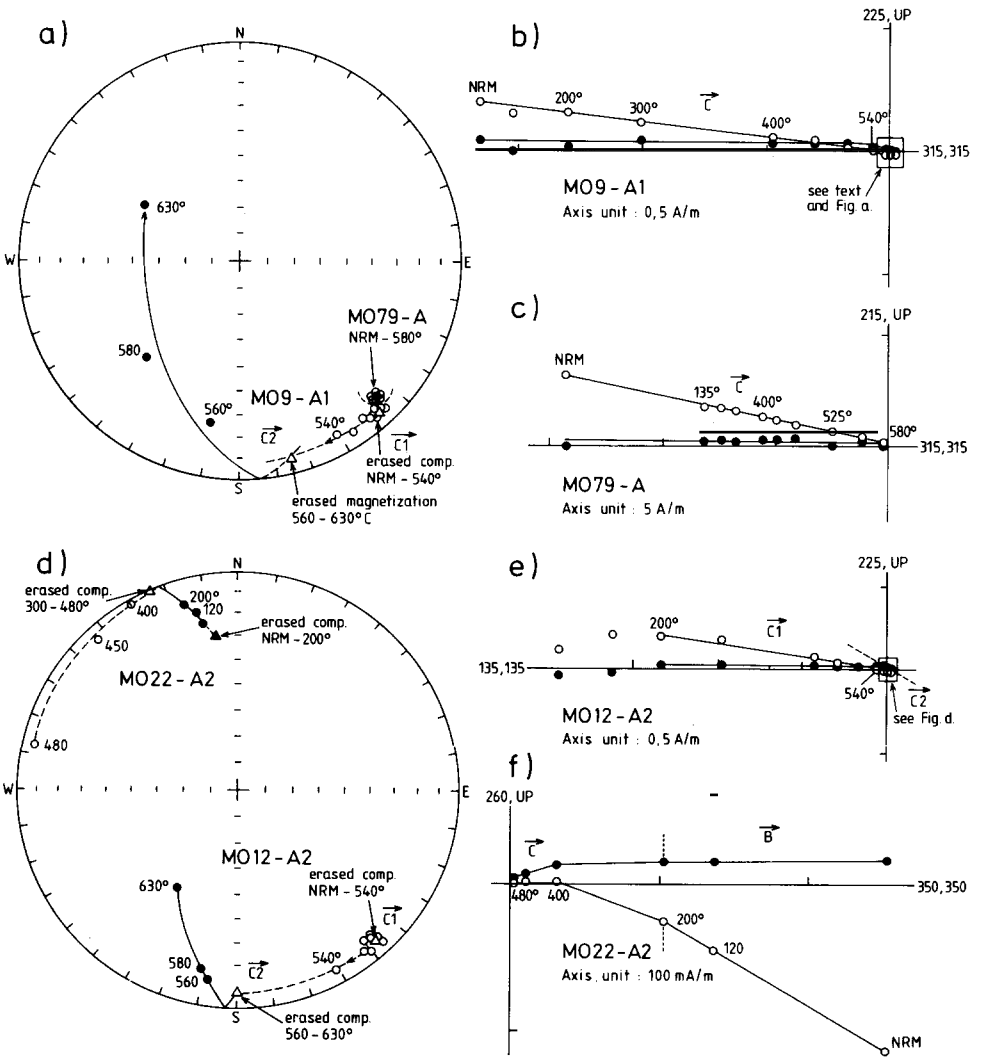


Fig. 6. Demagnetization results from the Monchique Complex, depicting largely coexisting B and C remanences. Samples of particular interest are those which change remagnetization trends in such a way that two C vectors (C1 and C2) can be defined. Diagram conventions as for Figs. 2 and 4.

TABLE 1

Isotopic age data from the Algarve Alkaline Igneous Province

Site sample (present study)	Rock type	Method	K_2O (wt. %)	Rad. ^{40}Ar ($mm^3\ g^{-1}$)	Atm. content (%)	Age [a] $\pm 1\sigma$ Ma	Paleo-magnetization group	References
Monchique Complex								
	Syenite	K–Ar, nepheline				74.2 ± 1.5		Macintyre and Berger (1982)
	Syenite	K–Ar, nepheline				69 ± 1.5	A	Macintyre and Berger (1982)
	Syenite	K–Ar, biotite				71.6 ± 1.5		Macintyre and Berger (1982)
Site 1	Syenite	K–Ar, w.r.	6.76 ± 0.05	$(1.49 \pm 0.01) \times 10^{-2}$	13.4	67.1 ± 0.7		Present study
6	Syenite	K–Ar, w.r.	6.77 ± 0.04	$(1.53 \pm 0.01) \times 10^{-2}$	10.2	68.7 ± 0.6		Present study
13	Basic segreg. in syenite	K–Ar, w.r.	2.83 ± 0.03	$(6.43 \pm 0.05) \times 10^{-3}$	20.5	69.1 ± 0.9		Present study
19	Lamprophyre dyke	K–Ar, w.r.	3.55 ± 0.04	$(7.71 \pm 0.05) \times 10^{-3}$	32.7	66.1 ± 0.9	B	Present study
	Syenite	K–Ar, feldspar				68 ± 1.4		Macintyre and Berger (1982)
	Syenite	K–Ar, feldspar				66.7 ± 1.4		Macintyre and Berger (1982)
7	Lamprophyre dyke	K–Ar, w.r.	3.15 ± 0.03	$(6.20 \pm 0.05) \times 10^{-3}$	12.1	60 ± 0.7	'C'	Present study
Salema intrusion								
Al 6	Volcanic plug	K–Ar, w.r.	1.01 ± 0.01	$(2.27 \pm 0.02) \times 10^{-3}$	21.2	68.3 ± 0.9		Present study
AL 12	Volcanic plug	K–Ar, w.r.	1.65 ± 0.02	$(3.82 \pm 0.03) \times 10^{-3}$	14.0	70.4 ± 1.0	B	Present study
AL 17	Volcanic plug	K–Ar, w.r.	0.84 ± 0.007	$(1.94 \pm 0.014) \times 10^{-3}$	21.4	69.8 ± 0.8		Present study
AL 7	Volcanic plug	K–Ar, w.r.	1.03 ± 0.01	$(2.19 \pm 0.02) \times 10^{-3}$	23.6	64.7 ± 0.9	'C'	Present study
AL20	Volcanic plug	K–Ar, w.r.	1.20 ± 0.01	$(2.42 \pm 0.02) \times 10^{-3}$	17.3	61.5 ± 0.7		Present study

w.r., Whole-rock analysis.

[a] $\lambda_e = 0.581 \times 10^{-10}\ yr^{-1}$; $\lambda_\beta = 4.962 \times 10^{-1}\ yr^{-1}$; $^{40}K/K = 1.167 \times 10^{-2}$ at. % (Steiger and Jäger, 1977).

Monchique magnetizations. However, in cases of coexisting A and B magnetizations it is generally the A component that has the higher AF or thermal stability. The A component is found in all rock types, militating against the possibility that the dyke swarm might represent a significant later magmatic event responsible for superimposing either the B or the C magnetization.

4. Radiometric ages and their interpretation

Conventional K–Ar analysis has been carried out on a total of 10 samples. These data are included in Table 1, which gives the calculated mean age with 1 S.D. error based on the analytical reproducibility.

Of the five Monchique samples investigated two were syenites (sites 1 and 6), one came from a basic granular rock (site 13), and the last two were from lamprophyre dykes (sites 7 and 19). Apart from the discrepantly low age of 60.0 ± 0.7 Ma from site 7, the remaining four ages are indistinguishable at the 2σ confidence level and are furthermore concordant with two feldspar ages obtained by Macintyre and Berger (1982). The unweighted mean age of these six samples is 67.6 $\pm$ 1.2 Ma. We present also five age determinations on samples from the previously undated Salema intrusion. Three of these results are indistinguishable from the Monchique dates at the 2σ confidence level, whereas two samples yielded lower ages which are believed to be indicative of argon loss. Macintyre and Berger (1982) demonstrated that the feldspar ages (which coincide with the bulk of the whole-rock ages of the present study, were systematically lower (~ 5 Ma) than those obtained from coexisting biotite and nepheline, minerals regarded as fairly reliable igneous chronometers (Mcintyre et al., 1969). They concluded from the evidence of their mineral ages (Table 1) that the Monchique pluton was emplaced in the interval 72 ± 2 Ma. This age is consistent with the palaeomagnetic evidence and is regarded here as corresponding to the principal A component of the Monchique magnetization. The four Monchique whole-rock ages, two feldspar ages of Macintyre and Berger (1982), and three Salema results which form a concordant group yield a mean age of 68 ± 2 Ma. The fact that the Salema intrusion (unlike the Monchique pluton) is devoid of magnetic overprinting leads us to conclude that the 68 ± 2 Ma age corresponds to the time of emplacement of the Salema intrusion. The Salema emplacement event did, however, cause penetrative magnetic and radiometric overprinting of the Monchique Complex. Although the palaeomagnetic resetting, expressed in the B component, was of moderate importance (leaving a reasonable portion of the original A magnetization unaffected), the whole-rock K–Ar system was apparently completely reset (reflecting the relatively poor argon retentivity of feldspar).

The two young ages obtained on Salema material, 61.5 ± 0.7 and 64.7 ± 0.9 Ma, may be a consequence of a later thermal (–chemical) event. The Salema intrusion is transected by a number of faults along which bleaching and other visible alteration effects are clearly recognized. In addition to the 60.0 ± 0.7 Ma age obtained from a Monchique dyke (site 7), K–Ar analysis in the Sines Complex yielded two radiometric dates in this age range, 62.0 ± 1.3 and 63.8 ± 0.8 Ma (Storetvedt et al., 1987). It is likely that these younger ages are not true rock ages but represent mixed dates due to partial argon loss during a late phase of reactivation of the fault zone regarded as the tectonomagmatic link between the Sintra, Sines and Algarve plutonic provinces (Ribeiro, 1979). This explanation accords with the inferred late Palaeocene–Eocene age for the shallow NNW–SSE-directed magnetizations recorded by the Algarve rocks (see below).

5. Origin of the palaeomagnetic pattern

From the discussion above we conclude that the Monchique A component represents the oldest magnetization reported in the present study, having most likely been acquired at the time the complex cooled 72 ± 2 Ma ago. It is further inferred that the Salema intrusion received its magnetization when it cooled 68 ± 2 Ma ago. This latter thermal event in the Algarve region led to

magnetic and radiometric overprinting in the Monchique Complex.

Establishing the age of the Burgau magnetization, which corresponds to the C component of the Monchique Complex, is more problematic. Because of the evident alteration of the limited quantity of material obtained from this intrusion, K–Ar dating has not been attempted. However, the Burgau magnetization ($D = 333°$, $I = 11°$) is indistinguishable from an inferred overprinted component in the Lisbon volcanics ($D = 333°$, $I = 14°$) apparently carried by maghaemite (Storetvedt et al., 1987). The latter evidence, together with the oxidized nature of the Burgau rocks, and the relatively sporadic occurrence of the corresponding Monchique magnetization (the C component), favours a low-temperature (secondary) origin for this shallow NNW–SSE-directed magnetization. This conclusion is further substantiated by the fact that our C remanence characterizes lower Tertiary rocks of the Ebro Basin and the Catalan Ranges (Pares et al., 1988). Also, Storetvedt (1990) has shown that magnetizations corresponding to the C component have a very wide geographic distribution, and that this palaeomagnetic field was in existence from late Palaeocene to around the Eocene–Oligocene boundary.

Table 2 summarizes the overall palaeomagnetic directions for individual magnetizations of the Algarve Alkaline Igneous Province. These data, with those previously published (Storetvedt et al., 1987) for the Sintra and Sines complexes (see Fig. 1a for locations), permit a full re-evaluation of the plate tectonic setting of Iberia. Before developing this potentially controversial discussion, it is necessary to examine the wider perspective of Iberian palaeomagnetism.

5.1. The overall significance of Iberian palaeomagnetic data

Figure 7 displays some recently determined palaeopoles of Permo-Triassic and late Cretaceous–Tertiary rocks for Iberia in conjunction with data for Africa, Europe and the islands off NW Africa. The suggested Upper Cretaceous–Tertiary Apparent Polar Wander (APW) path for Africa (Fig. 7b) and the polar curve fitted through the Iberian data are strikingly similar. Furthermore, the Iberian results agree with the pattern portrayed by the late Mesozoic–Tertiary poles from the Canary and Cape Verde Islands (Fig. 7a). This implies that, at least from the uppermost Cretaceous onwards, the palaeomagnetic reference frame for Iberia is identical to that of the African

TABLE 2

Summary of palaeomagnetic results from the Algarve Igneous Province

Formation	Inferred magnetic age (Ma)	N	D_m	I_m	α_{95}	R	K
Monchique syenite							
Component A	72±2	27	181	−42	4.5	26.3	39.4
Component B	68±2	14	347	35	6.9	13.6	33.6
Component C	Lower Tertiary	8	138	−8			
Salema intrusion	68±2	70	341	42	1.7	69.3	93.7
Burgau intrusion, secondary component	Lower Tertiary	7	333	11	2.3 .	6.9	545.2

All group parameters are based on unit weight of specimens. The Monchique syenite has suffered magnetic overprinting at the time of the emplacement of the Salema intrusion at ~68 Ma (magnetization B), and in the Lower Tertiary (magnetization C). Statistical parameters have not been attached to the latter remanence because of its smeared distribution between stages of the relative Lower Tertiary field pattern. N, Unit specimen vectors; D_m, mean declination; I_m, mean inclination; α_{95}, half-angle of the cone of 95% confidence about the mean (in degrees); R, length of resultant vector; K, precision parameter. Statistics according to Fisher (1953).

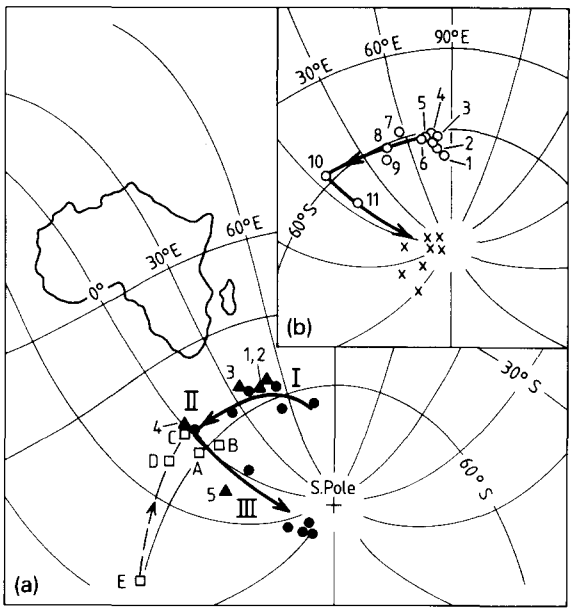


Fig. 7. (a) Polar pattern deduced from studies of the older volcanic complexes of the Canary and Cape Verde Islands, closed circles (Storetvedt et al., 1979; Storetvedt, 1980; Storetvedt and Løvlie, 1983; Abranches et al., 1990) and various rock formations of Iberia, numbered triangles. The Iberian data are as follows; 1, 2, Permo-Triassic rocks of the Cantabrian range (Van den Berg, 1980), 3, Lisbon volcanics, overprinted component (Storetvedt et al., 1987); 4, rocks of the Catalan Coastal Range (Pares et al., 1988); 5, Eocene rocks from the Ebro Basin (Pares et al., 1988). For comparison, some European results, square symbols, of Tertiary age, A–C (Løvlie et al., 1972; Løvlie and Kvingedal, 1975; Storetvedt, 1978), and of inferred Meso-Cenozoic age, D and E (Storetvedt et al., 1978); Storetvedt and Carmichael, 1979; Storetvedt, 1987), are also displayed. The inset diagram (b) shows palaeomagnetic poles relative to continental Africa in which crosses represent late Tertiary data, and the numbered poles are from Mesozoic to Lower Tertiary rocks as follows: 1, Mlanje (Gough and Opdyke, 1963; Briden, 1967); 2, mean Mesozoic SE Africa (Hailwood and Mitchell, 1971); 3, Lupata, 106 Ma (Gough and Opdyke, 1963; Gough et al., 1964); 4, Shawa ijolite (Gough and Brock, 1964); 5, mean Mesozoic NW Africa (Hailwood and Mitchell, 1971); 6, Hoachanas (Gidskehaug et al., 1975); 7, Kimberlite pipe, ~86 Ma (McFadden and Jones, 1977); 8, Kimberlite pipes, 90 Ma (Hargraves and Onstott, 1980); 9, Wadi Natash volcanics, 81–90 Ma (El Shazly and Krs, 1973); 10 and 11, Lower (Middle) Tertiary volcanics Egypt (Gouda Hussain et al., 1979). The very close match in the APW patterns of the data sets for Africa, Iberia and Canary–Cape Verde Islands is noteworthy. Stages I, II and III of the polar trend refer to corresponding stages in the directional changes depicted in Fig. 8. Poles 1–5 correspond to directions 1–5 of Fig. 8.

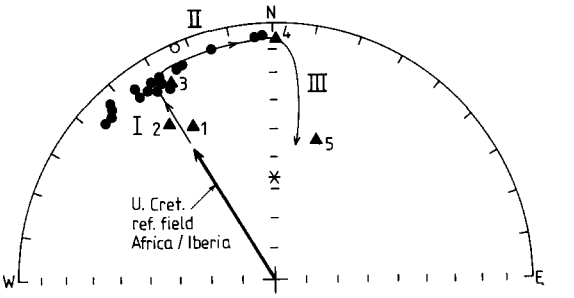


Fig. 8. Mean palaeomagnetic directions of Iberian rock, numbered triangles, corresponding to those outlined in Fig. 7, in association with the C group (individual specimen results) of the Monchique Complex, and the Burgau results. The cross represents the direction of the present axial geocentric dipole field for central Iberia. Stages of relative field changes (I–III) correspond to those of Fig. 7. Other conventions as for Fig. 2.

plate. To avoid confusing the diagram, the Iberian data referred to in Fig. 7 are restricted to those displayed in Fig. 8. The vast majority of results from Permian–late Cretaceous rocks of the Peninsula are, however, distributed close to the Upper Cretaceous–Lower Tertiary westward-trending polar branch, with a concentration in the longitude range of 20–60° E (see Piper, 1988, for complete data listing). The Permo-Carboniferous data depict anomalously high palaeolatitudes, which cannot be reconciled with the late Palaeozoic latitudinal frames of the adjacent Africa and Europe plates. However, the fact that the great majority of Iberian palaeopoles from late Palaeozoic and Mesozoic rocks fit so well with the late Cretaceous–early Tertiary APW curve for Africa leads us to conclude that the relative rotational movement between Africa and Europe in late Cretaceous and early Tertiary time (see also below) provided thermo-chemical agents for extensive magnetic resetting of Iberian rock formations. Also, the bulk of Iberian palaeomagnetic studies is in rock formations from the northern and eastern regions of the Peninsula, areas that were particularly affected by Alpine orogenic deformation, which give further substance to the remagnetization hypothesis. On the contrary, the western sector of the Peninsula, including Portugal, was shielded from significant Alpine diastrophism, so that its detailed Upper Cretaceous palaeomagnetic

pattern should therefore have remained fairly intact against subsequent magnetic resetting.

5.2. *The late Mesozoic–early Tertiary palaeofield; comparison of Africa and Europe*

Because of lack of suitable rocks, reliable European palaeopoles of Jurassic and Cretaceous age are very difficult to find. For comparison with the African and Iberian data we have therefore plotted two well-defined overprinted poles from Old Red Sandstone rocks which lie adjacent to the Great Glen Fault (north Scotland), inferred to have been impressed during Alpine (Meso-Cainozoic) reactivation of the Fault (Storetvedt, 1987). These results when combined with Lower Tertiary poles for Europe (Løvlie et al., 1972; Løvlie and Kvingedal, 1975; Storetvedt, 1978) apparently define an eastward-migrating polar trend for Europe joining the westward-trending APW curve for Africa at around 55°S, 0°E. Faced with the evidence of Fig. 7, we conclude that Africa and Europe underwent a relative rotation in late Cretaceous–early Tertiary time (see also below).

Subtropical conditions in Europe during the early Tertiary have long been recognized from palaeontological evidence (Wolfe, 1980). The final polar wander branch for Africa and Europe, running roughly along the Greenwich meridian from ~ 55°S to the present pole, may be a consequence of a rapid polar movement which occurred at around the Eocene–Oligocene boundary. Support for this proposition comes from floral and faunal evidence (Wolfe, 1978; Collinson et al., 1981; Pomerol, 1982) which indicates a substantial temperature decline in the northern hemisphere late in the Eocene. This palaeontological evidence may also lead to the conclusion that the APW segment that runs close to the 60°S parallel (Fig. 7) are at least partly of early Tertiary age. The Upper Cretaceous–Lower Tertiary APW paths for Africa and Europe are further substantiated by Storetvedt (1990).

5.3. *The palaeomeridian discrepancy between Africa and Europe; constraints on relative plate motion*

Based on the well-established discrepancy between the Permian–early Mesozoic palaeomeridians for Africa and Europe, a ~ 50° relative rotation between the two continental blocks must have taken place. According to the arguments above, it seems that all of this rotation might have occurred in late Cretaceous to early Tertiary time. However, the evidence of geological continuity across the major crustal fractures between Morocco and France requires that during the relative Africa–Europe plate rotation, Iberia did not undergo extensive independent east–west translations with respect to the two continents. These geological constraints can be satisfied within a kinematic model that includes clockwise rotation of Europe and anticlockwise rotation of Africa. This model accounts for the fact that the plate boundary zone, which comprises the Mediterranean and Alpine regions, consists of a complex mobile belt which displays major fractures, tectonic deformation and micro-plate rotations. Depending on the instantaneous speed of rotation of the major and minor plates involved, both dextral and sinistral displacements may have occurred along the various faults of the mobile belt, so that individual micro-plates (Iberia for example) could have undergone both clockwise and anticlockwise rotations.

5.4. *Compilation of Cretaceous–Tertiary igneous palaeomagnetic data for Portugal; tectonic implications*

Palaeomagnetic and isotopic age evidence from the alkaline igneous rocks of Portugal indicate that the ambient geomagnetic field inclination in Campanian–Maastrichtian time was ~ 41°, and data depicted in Figs. 7 and 8 suggest that the relative palaeomeridian direction at that time was northeasterly (~ 330°). This declination, which also coincides with that of the African palaeofield at around the K–T boundary has therefore been adopted as the appropriate palaeomagnetic reference direction for Iberia in evaluating its microplate rotations. Figure 9 summarizes the various magnetization vectors derived from late Cretaceous igneous rocks of Portugal. These results, with their associated radiometric ages, constitute a unique set of data that cannot be reconciled with the prevailing plate tectonic model for Iberia which

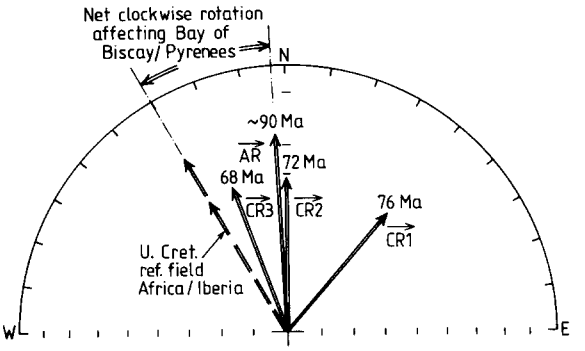


Fig. 9. The various palaeomagnetic vectors derived from alkaline intrusives of Portugal: CR1, Sines–Cabo da Roca (Storetvedt et al., 1987); CR2, Monchique primary magnetization (component A, this paper); CR3, Salema intrusion (this paper); AR, Sintra pluton (Storetvedt et al., 1987). The reference field with an approximate declination of 330° (see text) had an inclination of about 41° in the ~ 76–65 Ma time-span (Campanian–Maastrichtian), but at around 90 Ma (Turonian) the inclination was apparently shallower (cf. inclination of the Sintra magnetization, AR). It is assumed that all magnetic vectors depicted here were originally aligned along the ambient field directions (declination ~ 330°), i.e. their present declination discrepancies must have tectonic causes. The progressive swing vs age of the CR vectors demonstrates very clearly a clockwise rotation of ~ 70°, and the position of the AR vector gives evidence for a ~ 40° angular movement (the declination difference between the AR and CR1 vectors) in the counter-clockwise sense. The estimated net clockwise rotation, i.e. the angular difference between the AR vector and the reference field, probably applied to the whole of Iberia, providing the impetus for the tectonic deformation in the Bay of Biscay and the Pyrenees. Diagram conventions as for Fig. 2.

includes a ~ 35° anticlockwise rotation with respect to an assumed European parental plate. In fact, all of the vectors in Fig. 9 are at variance with the Africa–Iberia reference field for late Cretaceous–early Tertiary time. The progressive anticlockwise swing, at systematically decreasing ages for the Campanian–Maastrichtian vectors (CR1–CR3), apparently 'docking' with the reference field close to the Cretaceous–Tertiary boundary, is particularly noteworthy. In the latest Cretaceous–early Tertiary reference plane (declination ~ 330°) a CR vector would correspond to a palaeomagnetic pole at ~ 60°S, 61°E. As the corresponding polar trend appears to be progressively younging westward (cf. Figs. 7 and 8), we have to conclude therefore that a major part of this polar branch is of Lower Tertiary age. In tectonic terms the latter sequence of palaeomagnetic vectors express a clockwise rotation of ~ 70°. The position of the older AR magnetization (Sintra Complex) implies, however, that the clockwise movement was preceded by a counter-clockwise rotation of at least 40°. Rotations of this magnitude must have been linked to major tectonic deformation in the Iberian domain. The fact that the Alpine diastrophism had its maximum intensity in northern and southeasterly regions of the Peninsula (the western areas, including Portugal, were only marginally affected) leads us to believe that the rotations operated over the whole of the Iberian Peninsula, and that the various igneous complexes of SW Portugal were injected at different times during a period of regional stress–tectonic instability. The net clockwise rotation of the order of 30° (70–40°) could to a major extent have been responsible for the strong orogenic pulse that affected the Peninsula in uppermost Cretaceous and Lower Tertiary time. The net clockwise rotation would be the 'correction' figure that applies to Iberian palaeomagnetic data predating the Alpine kinematic events.

The consideration above suggests that Iberia's independent kinematic behaviour is confined to the initial phase of the relative movement between Africa and Europe in the Upper Cretaceous. The significant part of the relative Africa–Europe rotation appears to have taken place in the Lower Tertiary, during which Iberia and Africa moved as a united plate. The smeared distribution of the shallow secondary magnetization found in the Portuguese alkaline igneous rocks (cf. Fig. 8), consistent with the characteristic magnetizations of some Lower Tertiary rocks of NE Spain (Pares et al., 1988), defines the final anticlockwise rotation of Africa–Iberia. This latter rotation is unrelated to the suggested anticlockwise rotation of Iberia previously thought to be responsible for the opening-up of the Bay of Biscay.

Our suggested oldest magnetization, the AR vector (main Sintra pluton), is the least satisfactory element in our model. Thus, its age is at present not precisely established (see also below). Furthermore, the characteristic magnetization is somewhat shallower than that displayed by the

other alkaline intrusive bodies investigated. The original Sintra magnetization direction reported, $D = 359°$, $I = 43°$ ($\alpha_{95} = 5°$) (Van der Voo, 1969), would as far as inclination is concerned, be consistent with that of the other intrusives ($I = 41°$). However, we subsequently found that the granitic rocks of the pluton were palaeomagnetically unreliable, because of their relatively large component of recent chemical (weathering) magnetization (Storetvedt et al., 1987). The shallower characteristic remanence, $D = 358°$, $I = 27°$ ($\alpha_{95} = 3°$), was established from younger intrusive phases of gabbros and diorites. However, the intrusion mechanism of the Sintra pluton is associated with a tectonic component (Ribeiro, 1979), so at present we cannot rule out a tectonic explanation for the shallower inclination. These age and magnetization aspects will have to be settled in future studies, but possible minor modifications of the present inclination for Sintra will not influence the general tenets of our kinematic model.

6. Plate tectonic model

According to the palaeomagnetic arguments advanced above, relative plate rotation between Africa and Europe began some 100 Ma ago (Cenomanian). Africa moved in a counter-clockwise sense whereas Europe performed a clockwise rotation, creating a belt of extensive tectonic deformation in the intervening boundary zone (Alpine–Mediterranean region). Despite the presence of major fracture zones which bound the Iberia plate (Azores–Gibraltar Fracture Zone and North Pyrenean Fault Zone), the Peninsula does not seem to have undergone any extensive independent east–west translation with respect to adjoining Africa or Europe. However, its tectonically unstable position, trapped between the two major plates, gave rise to two consecutive rotations which were opposite in sense.

The initial phase of micro-plate rotation was of at least 40° in a counter-clockwise sense. An upper age bound of ~ 76 Ma can be set for this kinematic process (Lower Campanian), but a lower age limit cannot be precisely defined at present. Thus, the ~ 90 Ma age for the main Sintra pluton is not well established, so older isotope ages in the 90–100 Ma range, determined from alkaline magmatic rocks in the Pyrenees (Montigny et al., 1986), may perhaps be a more relevant date for the onset of the anticlockwise rotation. The anticlockwise movement was followed by a clockwise rotation of ~ 70°, which can be temporally well constrained at between ~ 76 and ~ 65 Ma. Thus, the period of pronounced plate tectonic instability for Iberia ended at around the Cretaceous–Tertiary boundary. Relative rotation between Africa and Europe continued into the Lower Tertiary, but throughout that time Iberia's rotation was linked to that of Africa.

The kinematic model presented here is capable of explaining the postulated tectonic evolution of the Bay of Biscay and the Pyrenees. According to our model, the generally accepted divergent motion in the Bay of Biscay in the Middle–Upper Cretaceous, possibly associated with strike-slip movement along the North Pyrenean Fracture system (Choukroune and Mattauer, 1978; Olivet et al., 1982; Boillot, 1984) would be, at least in part, a consequence of the postulated anticlockwise rotation. Also, the early Alpine compressive phase in the Betic Cordillera (SE Spain) was probably strongly linked to this counter-clockwise movement, and the Betic region is likely to have undergone remagnetization in this process. The subsequent clockwise rotation, which provided a net crustal shortening along the northern boundary of the Peninsula, would be strongly responsible for the Pyrenean orogeny as well as for the compressive deformation occurring in the Bay of Biscay. Evidence for such a compressive phase exists in data from the southern end of the ECORS Aquitaine reflection profile (Pinet et al., 1987), where features such as underthrusting, important Moho deepening and thickening of the high-velocity lower crust have been identified. The 'anomalous' NE–SW declinations characterizing Mesozoic rocks of the Betic Cordillera (Osete et al., 1988) are also readily explained by the late Cretaceous clockwise rotation.

The evidence that an important part of the relative Africa–Europe plate movement took place in the Lower Tertiary (i.e. after the independent tectonic rotations of Iberia had ceased), at a time

when Iberia was moving with Africa, implies that the North Pyrenean Fracture Zone must have formed a primary plate boundary along which transcurrent movement would have occurred. The steeply dipping North Pyrenean Fault and the associated Moho offset (Daigniers et al., 1982) could be a consequence of lateral movement along this fracture zone in the Lower Tertiary, so that evidence for the late Cretaceous deep crustal deformation (following the net clockwise rotation of Iberia) might eventually be found either in the eastward or westward extension of the Pyrenean chain. In the tectonic model we have presented the Pyrenean orogeny may be explained partly as the result of crustal shortening, following the net clockwise rotation of Iberia, and partly as a consequence of transpression caused by shear motions along the North Pyrenean Fault Zone. The palaeomagnetic evidence shows that the most important period of deformation was from Campanian to well into the Lower Tertiary, a conclusion that is fully consistent with the available geological evidence. Finally, our tectonic model refutes earlier proposals that Iberia performed a general counter-clockwise rotation in Middle–Upper Cretaceous time, creating the Bay of Biscay. We regard the earlier model as an artefact which arises from an incorrectly identified palaeomagnetic reference frame, which linked Iberia to Europe instead of Africa. In fact, the total movement of Iberia was clockwise, resulting in a net crustal shortening in the Bay of Biscay–Pyrenean region. Figure 10 summarizes the main kinematic history.

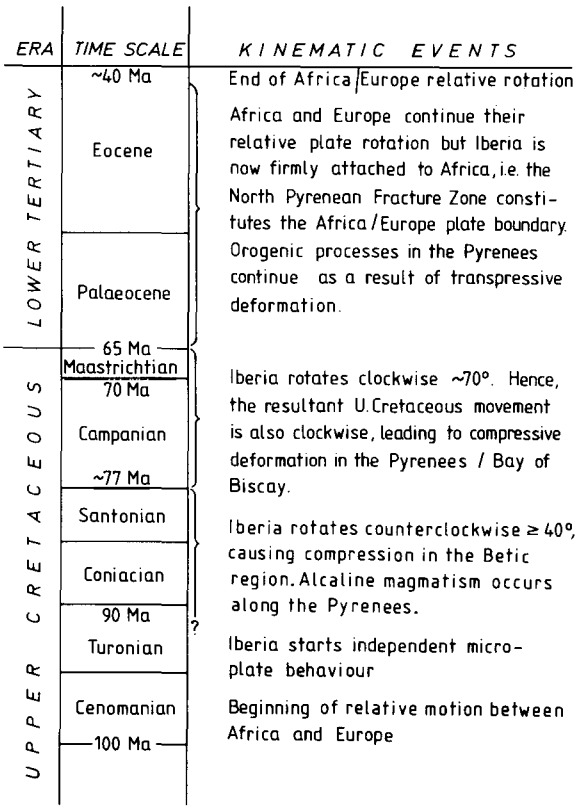


Fig. 10. The main Alpine kinematic events for Iberia.

Continental attenuation processes must therefore be invoked to explain the crustal evolution of the Bay of Biscay. From late Jurassic times, and throughout the Lower Cretaceous, pronounced rifting and subsidence occurred in the northern continental margin of the Bay (de Charpal et al., 1978) as well as in the Aquitanian Basin (Brunet, 1984), and the ECORS reflection profile demonstrates extensive crustal thinning beneath the Parentis Basin (Pinet et al., 1887). In fact, the abruptly disappearing southward-dipping reflectors at the 'base' of the thinned crust on the northern flank of the Parentis Basin are strongly suggestive of a mechanism of crustal loss to the upper mantle (Pinet et al., 1987).

Acknowledgements

This work was supported by the Instituto Nacional de Investigaçao Cientifica (INIC). Financial support to M.C. Abranches and K.M. Storetvedt through the cultural treaty between Portugal and Norway is gratefully acknowledged. We thank Dr. J. Munha for guidance in the field areas, Knut Breyholtz for sample preparation, Rob Ridley for analytical support, Dorothy Cooper for the diagrams, and Maureen Hopkinson and Elin Jarland Eriksen for typing the manuscript. We convey also our thanks to an anonymous reviewer for constructive comments.

References

Abranches, M.C., Storetvedt, K.M., Serralheiro, A. and Løvlie, R., 1990. The palaeomagnetic record of the Santiago volanics (Rep. of Cape Verdo); multiphase magnetization and age consideration. Phys. Earth Planet. Inter., in press.

Assunçao, C.F.T., 1940. Sobre e petrologia das formacoes eruptivas do Algarve. Bol. Mus. Lab. Min. Geol. Univ. Lisboa, 3rd Ser., 7–8: 47–81.

Boillot, G., 1984. Some remarks on the continental margins in the Aquitaine and French Pyrenees. Geol. Mag., 121: 407–412.

Briden, J.C., 1967. A new palaeomagnetic result from the Lower Cretaceous of East Central Africa. Geophys. J.R. Astron. Soc., 12: 375–380.

Brunet, M.-F., 1984. Subsidence history of the Aquitaine basin determined from subsidence curves. Geol. Mag., 121: 421–428.

Choukroune, P. and Mattauer, M., 1987. Tectonique des plaques et Pyrénées: sur le fonctionnement de la faille transformante nord-pyrénéenne; comparaisons avec des modèles actuels. Bull. Soc. Géol. Fr., 20: 689–700.

Collinson, M.E., Fowler, K. and Boulter, M.C., 1981. Floristic changes indicate a cooling climate in the Eocene of southern England. Nature, 291: 315–317.

Daigniers, M. Gallart, J., Banda, E. and Hirn, A., 1982. Implications of the seismic structure for the orogenic evolution of the Pyrenean Range. Earth Planet. Sci. Lett., 57: 88–100.

De Charpal, O., Guennoc, P., Montadert, L. and Roberts, D.G., 1978. Rifting, crustal attenuation and subsidence in the Bay of Biscay. Nature, 275: 706–711.

El Shazly, E.M. and Krs, M., 1973. Palaeogeography and palaeomagnetism of the Nubian Sandstone, Eastern Desert of Egypt. Geol. Rundschau, 62: 212–225.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, A, 217: 295–305.

Gidskehaug, A., Creer, K.M. and Mitchell, J.G., 1975. Palaeomagnetism and K–Ar ages of the South-West African basalts and their bearing on the time of initial rifting of the South Atlantic Ocean. Geophys. J.R. Astron. Soc., 42: 1–20.

Goncalves, F., 1967. Subsidios para o conhecimento geologico do macico eruptivo de Monchique. Com. Serv. Geol. Portugal, 52: 169–184.

Gouda Hussain, A., Schult, A. and Soffel, H., 1979. Palaeomagnetism of the basalts of Wadi Abu Tereifiya, Mandisha and diorite dykes of Wadi Abu Shihaf, Egypt. Geophys. J.R. Astron. Soc., 56: 55–61.

Gough, D.I. and Brock, A., 1964. The palaeomagnetism of the Shawa ijolite. J. Geophys. Res., 69: 2489–2493.

Gough, D.I. and Opdyke, N.D., 1963. The palaeomagnetism of the Lupata alkaline volcanics. Geophys. J.R. Astron. Soc., 7: 457–468.

Gough, D.I., Brock, A., Jones, D.L. and Opdyke, N.D., 1964. The palaeomagnetism of the ring complexes at Marangudzi and the Mateke Hills. J. Geophys. Res., 69: 2499–2507.

Hailwood, E.A. and Mitchell, J.G., 1971. Palaeomagnetic and radiometric dating results from Jurassic intrusions in South Morocco. Geophys. J.R. Astron. Soc., 24: 351–364.

Hargraves, R.B. and Onstott, T.C., 1980. Palaeomagnetic results from some southern African kimberlites, and their tectonic signficance. J. Geophys. Res., 85: 3587–3596.

Løvlie, R. and Kvingedal, M., 1975. A palaeomagnetic discordance between a lava sequence and an associated interbasaltic horizon from the Faeroe Islands. Geophys. J.R. Astron. Soc., 40: 45–54.

Løvlie, R., Gidskehaug, A. and Storetvedt, K.M., 1972. On the magnetization history of the Northern Irish Basalts. Geophys. J.R. Astron. Soc., 27: 487–498.

Macintyre, R.M. and Berger, G.W., 1982. A note on the geochronology of the Iberian Alkaline Province. Lithos, 15: 133–136.

Macintyre, R.M., York, D. and Gittins, J., 1969. The K–Ar characteristics of nepheline. Earth Planet. Sci. Lett., 7: 125–131.

McFadden, P. and Jones, D.L., 1977. The palaeomagnetism of some Upper Cretaceous kimberlite occurrences. Earth Planet. Sci. Lett., 34: 125–135.

Montigny, R., Azambre, B., Rossy, M. and Thuizat, R., 1986. K–Ar study of Cretaceous magmatism in the Pyrenees: age and length of rotation of the Iberian Peninsula. Tectonophysics, 129: 257–273.

Olivet, J.L., Bonnin, J., Beuzart, P. and Auzande, J.M., 1982. Cenématique des plaques et paléogéographie: une revue. Bull. Soc. Géol. Fr., 24 (7): 875–892.

Osete, M.L., Freeman, R. and Vegas, R., 1988. Preliminary palaeomagnetic results from the Subbetic Zone (Betic Cordillera, southern Spain): kinematic and structural implications. Phys. Earth Planet. Inter., 52: 283–300.

Pares, J.M., Banda, E. and Santenach, P., 1988. Palaeomagnetic results from the southeastern margin of the Ebro Basin (northeastern Spain): evidence for a Tertiary clockwise rotation. Phys. Earth Planet. Inter., 52: 267–282.

Pinet, B., Montadert, L., Curnelle, R., Cazes, M., Marillier, F., Rolet, J., Tomassino, A., Galdeano, A., Patriat, Ph., Brun, M.F., Olivet, J.L., Schaming, M., Lefort, J.P., Arrieta, A. and Riaza, C., 1987. Crustal thinning on the Aquitaine shelf, Bay of Biscay, from deep seismic data. Nature, 325: 513–516.

Piper, J.D.A., 1988. Palaeomagnetic Database. Open University Press, Milton Keynes, 264 pp.

Pomerol, C., 1982. The Cenozoic Era: Tertiary and Quaternary. Ellis Norwood, Chichester, 272 pp.

Ribeiro, A., 1979. In: A. Ribeiro et al. (Editors), Introduction á la Géologie Génerale du Portugal. Serv. Geol. Portugal, Lisboa.

Rock, N.M.S., 1982. The late Cretaceous Alkaline Igneous Province in the Iberian Peninsula, and its tectonic significance. Lithos, 15: 111–131.

Steiger, R.H. and Jäger, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett., 36: 359–362.

Storetvedt, K.M., 1978. Structure of remanent magnetization in some Skye lavas, NW Scotland. Phys. Earth Planet. Inter., 16: 45–58.

Storetvedt, K.M., 1980. Fuerteventura palaeomagnetism and the evolution of the continental margin off Morocco. Phys. Earth Planet. Inter., 21: P1–P6.

Storetvedt, K.M., 1987. Major late Caledonian and Hercynian shear movements on the Great Glen fault. Tectonophysics, 143: 253–267.

Storetvedt, K.M., 1990. The Tethys Sea and the Alpine–Himalayan orogenic belt; mega-elements in a new global tectonic system. Phys. Earth Planet. Inter., 62: 141–184.

Storetvedt, K.M. and Carmichael, C.M., 1979. Resolution of superimposed magnetizations in the Devonian John O'Groats Sandstone, North Scotland. Geophys. J.R. Astron. Soc., 58: 769–784.

Storetvedt, K.M. and Løvlie, R., 1983. Magnetization properties of intrusive/extrusive rocks from East Maio (Rep. of Cape Verde) and their geological implications. Geophys. J.R. Astron. Soc., 73: 197–212.

Storetvedt, K.M., Carmichael, C.M., Hayatsu, A. and Palmer, H.C., 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic rock, NE Scotland: superimposed magnetizations, age of igneous activity, and tectonic implications. Phys. Earth Planet. Inter., 16: 379–392.

Storetvedt, K.M., Mogstad Våge, H., Aase, S. and Løvlie, R., 1979. Palaeomagnetism and the early magmatic history of Fuerteventura (Canary Islands). J. Geophys., 46: 319–334.

Storetvedt, K.M., Mogstad, H., Abranches, M.C., Mitchell, J.G. and Serralheiro, A., 1987. Palaeomagnetism and isotopic age data from Upper Cretaceous igneous rocks of W. Portugal; geological correlation and plate tectonic aspects. Geophys. J.R. Astron. Soc., 88: 241–263.

Van den Berg, J., 1980. New palaeomagnetic data from the Iberian Peninsula. Geol. Mijnbouw, 59: 49–60.

Van der Voo, R., 1969. Palaeomagnetic evidence for the rotation of the Iberian Peninsula. Tectonophysics, 7: 5–56.

Wilkinson, P., Mitchell, J.G., Cattermole, P.J. and Downie, C., 1986. Volcanic chronology of the Meru–Kilimanjaro region, Northern Tanzania. J. Geol. Soc. London, 143: 601–605.

Wolfe, J.A., 1978. A paleobotanical interpretation of Tertiary climates in the Northern Hemisphere. Am. Scientist, 66: 691–703.

Wolfe, J.A., 1980. Tertiary climates and floristic relationships at high latitudes in the Northern Hemisphere. Paleogeog., Palaeoclimatol., Palaeoecol., 30: 313–323.

Zijderveld, J.D.A., 1967. Demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and S.K. Runcorn et al. (Editors), Methods in Palaeomagnetism. Elsevier, New York, pp. 254–286.